The Guardians: serious game design for enhancing awareness of cultural heritage preservation

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Abstract: The Guardians is a serious game designed to be played on mobile devices for promoting awareness of cultural heritage. The Guardians consists of five stages. First, players can navigate in the game space and view brief information about heritage around the world. Second, once the player selects a specific cultural heritage he or she can learn background information about the site. Third, the player has to take quizzes to verify their understanding of the site. Fourth, the player can protect or repair endangered cultural properties in a mini game. Fifth, the game system allows players to donate to the actual cultural heritage that they have saved in the game. After playing the game, players around the world can communicate and form on-line communities. They can share their learning and cultural heritage - protecting simulation experiences - and cooperate to preserve the actual endangered heritage. Results of The Guardian's usability test showed that most players understood the mechanics and intentions of the serious game. Players commented that The Guardians enabled active learning, which is one of the important factors in increasing involvement and engagement in cultural heritage. Furthermore, players reported that increased awareness of the site eventually made them want to learn more about other sites and preserve cultural heritage more generally.

Keywords: cultural heritage; serious game; knowledge; preservation; awareness

1. Introduction

In this study a framework and prototype of a serious game, *The Guardians* was developed. The game was created to promote increased awareness of cultural heritage through gaming. The expected result of playing *The Guardians* was the promotion of a player's participation in cultural heritage preservation as a result of an increased awareness of its importance. Cultural objects are the basis of cultural memory which may be considered as an identity marker and distinguishing feature of a social or cultural group (Merryman 1989; Bessière 1998). Cultural heritage - including objects, monuments, inherited skills or symbolic representations around the world - is at risk due to weather, pollution, development and even tourism (Navrud & Ready 2002).

Without heritage, community identity is at risk; therefore preservation of cutural heritage is crucial. The one who protects endangered heritage is ultimately not a particular organization but an individual within a community. One of the best ways to preserve the legacy is to increase awareness of its importance. People's awareness is heightened when they know about the heritage, therefore educating the public about cultural heritage is important.

Some people feel that learning about heritage is tedious and protecting heritage is difficult or requires a lot of effort. This reserach suggests an easy and entertaining method of learning about cultural heritage for young adults. Serious games are a form of interactive computer-based games that have been developed to convey learning material (Deterding 2011). *The Guardians* is a serious game designed to be played on smart mobile devices.

According to Smartphone Ownership 2013 report,[1] 56 percent of American adults own a smartphone of some kind. As the report indicates, many people own mobile devices and research shows that mobile technology embedded learning offers learning in a natural environment (Schwabe & Goth, 2005). Therefore, playing serious games on mobile devices has the opportunity to provide effective learning. Furthermore, we have included a conceptual donation system in the game. According to Steinberg (1996), even for government registered National monuments, there is a tremendous shortage of funds for their upkeep and maintenance. We believe that learning about cultural heritage will increase people's awareness of the heritage which will affect people's willingness to donate and participate in endangered heritage preservation.

[1] PewResearch Center, Aaron Smith, Retrieved Oct 12, 2013 http://pewinternet.org/Reports/2013/Smartphone-Ownership-2013.aspx

2. Game Design

Conceptual Framework

Gameplay of *The Guardians* follows five stages. These stages illustrate a participation flow leading to cultural heritage preservation. The design goal of these stages was to educate players in natural and fun ways to heighten awareness of endangered heritage, eventually leading to active participation in preservation. Learning goals of playing the game can be evaluated by how much the players are willing to donate to preserve endangered heritage. Details and design rationales of each stage will be described in the following sections.

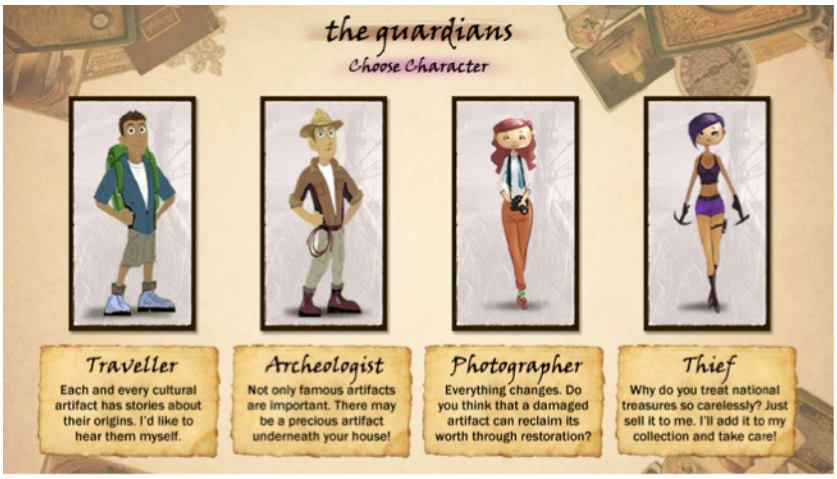
Image 2.1: Conceptual Framework for *The Guardians*



Stage 1. Interest and Discovery

This stage triggers player's interest in cultural heritage. The player can navigate in the game world and select an endangered heritage site or artifact of his or her interest. A good game scenario arouses deeper immersion (Park 2009). Immersive experience determines whether the player would continue to play or not (Chen 2007), therefore adapting good scenarios is also necessary for serious games. The scenario of the game is that there are four characters in a heritage preservation association, 'The Guardians', who protect heritage by participating in diverse missions. The game begins when the player selects one of four characters and becomes a member of The Guardians.

Image 2.2: Character Selection



Stage 2. Knowledge and Learning

Since *The Guardians* is a serious game, one of the game's main intentions is to educate players. This stage includes context and background information about endangered heritage. For this prototype, four endangered heritage sites registered by UNESCO are available.[2] See Table 2.1. According to Moreno (2001), learners in a group with a pedagogical agent remembered more contents and had higher problem solving transfer abilities. Inspired by this study, we used a non-player character (NPC) Paul, who serves as the pedagogical agent and suggests guidelines to motivate players. To reduce cognitive load, no more than four lines of information is displayed per page.

Image 2.3: Information Display by NPC



Paul, the NPC, delivers information about heritage in friendly colloquial ways.

Table 2.1: Four Endangered Heritage Sites in the Game

Name of Cultural Heritage	Country	Heritage Destruction Factor
Tombs of Buganda Kings at Kasubi (*Tombs)*	Kampala, capital of Uganda	Conflagration White ants
Mehmed Paša Sokolović Bridge in Viešgrad (*Bridge)*	Drina river, the border between Bosnia and Herzegovina and Serbia	Flood
Rio Pinturas (*Mural)*	Patagonia, Argentina,	Humidity (caused by visitors)
Arirang (*Folk Song)*	Korea	Absence of successor

[2] UNESCO (2013) World Heritage, from http://whc.unesco.org/en/about/, retrieved Oct 12, 2013.

Stage 3. Knowledge and Understanding

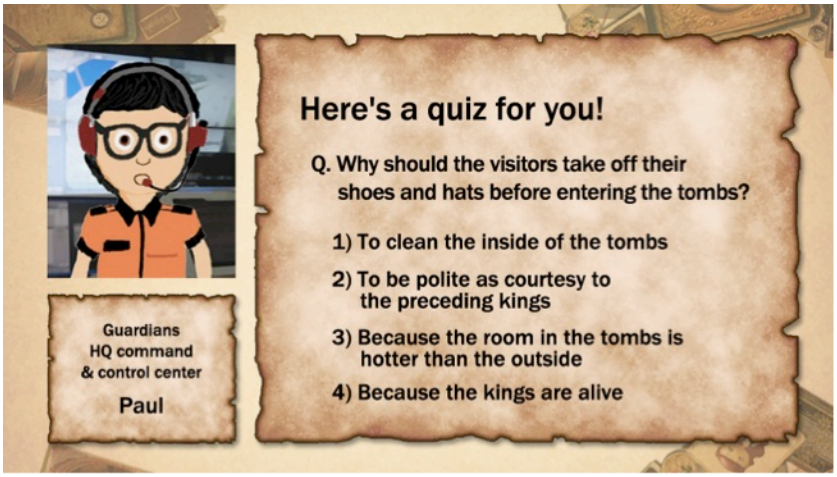
The players need to solve two levels of quizzes to verify whether they have understood the cultural heritage information they have learned. The quiz questions include general background information, destruction factors and protection methods. Correctly solving quizzes allows players to play mini-games. We have designed different levels of quizzes to promote knowledge gain and change attitudes towards cultural heritage, which are in the cognitive and affective domain of Bloom's taxonomy of learning (Bloom 1956). We have adopted Wang's knowledge-level and comprehension-level quizzes in our game (Wang 2008). The knowledge level quiz involves recollection and recognition of cultural heritage sites, while the comprehensive level emphasizes a player's understanding of the heritage to consider deeper meanings. An understanding of the heritage sites after learning is likely to promote awareness of the legacy.

Image 2.4: Knowledge Level (Tombs of Buganda Kings at Kasubi)



Items involve a learner's recall or recognition of ideas and concepts (Çepni 2006)

Image 2.5: Comprehensive Level (Tombs of Buganda Kings at Kasubi)



Items emphasize a learner's understanding of ideas and concepts (Çepni 2006)

Stage 4. Engagement

We chose to encourage learning through doing. The players can engage with the heritage by playing mini-games which are designed to eliminate heritage destruction factors. Refer to Table 2.2 for the directions and contexts of each game.

Table 2.2: Details of endangered heritage and mini-games

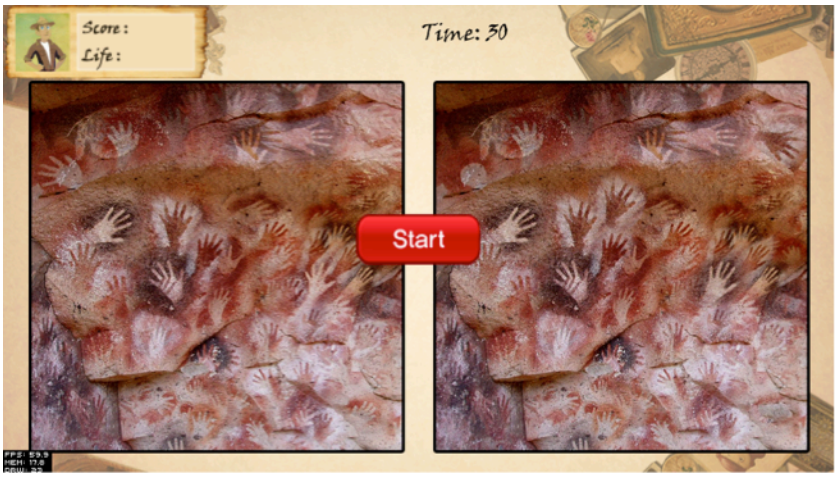
Name of Cultural Heritage	Game Mission & Activity
Tombs of Buganda Kings at Kasubi *(Tombs)*	Defend tombs from catching fire. Cut pouring fireballs in half before they hit the tombs and burn them down.
Mehmed Paša Sokolović Bridge in Viešgrad *(Bridge)*	Prevent the bridge from deluging. Quickly push button to stop water from overflowing.
Rio Pinturas *(Mural)*	The ancient painting was damaged by frequent exposure to the public. Detect deformed parts.
Arirang *(Folk Song)*	Succession of Arirang is at risk due to lack to succesors. Player has to choose a region of Arirang after listening to several Arirangs.

Image 2.6: Tombs of Buganda Kings at Kasubi Mini - game



The player needs to remove falling fireballs using the touch screen

Image 2.7: Rio Pinturas Mini-game



The left side is the original mural. The player needs to select five deformed parts of the mural within the time limit.

Stage 5. Participation

In the game, a conceptual donation system allows players to participate in the actual preservation of the heritage they have been 'protecting' in the game. The aim of the donation system is to extend heritage protection into the real world. Furthermore, a player's willingness to pay (WTP) is one possible reflection of the degree of a player's learning outcome and altered perspective.

3. Usability pilot test

Pilot test procedure and result

To test if the design intentions of *The Guardians* were effectively delivered, we conducted a pilot usability test using the fireball cutting game. Each participant watched the introduction, studied cultural heritage, took the quizzes, played the mini game, completed the survey, and had an in-depth interview. The entire process took about thirty minutes. Eight participants (3 males & 5 females) tried the prototype and conducted a usability and user experience survey through SUS (Bangor 2008). The results indicated a usability level grade of B+, which means that the players could play the game without confusion. During interviews, the players answered that learning about cultural heritage sites and playing the game made them understand more about those heritage sites and increased their interest in them.

Future work

Many participants mentioned that the textual information display should be more interactive and the usability of the game, especially the feedback system should be improved. These are possible areas for further development.

Image 3.1: Prototype Pilot Test



4. Conclusion

We designed a cultural heritage preservation serious game to raise people's awareness of our legacy. The learning system in the game develops knowledge and helps players to understand the importance of heritage preservation. Furthermore, the game system allows players to participate in actual preservation by donating. In the future, we will conduct an experiment to evaluate player's experiences on the entire game system and check whether game play had increased their awareness. Furthermore, the donation system is conceptual for this prototype, but we will modify the donation system to enable actual donation to endangered heritage preservation organizations for more active participation in cultural heritage preservation.

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